ASSIGNMENT – SED & AWK (Day-05)

1) What is Sed (Stream Editor)?

Sed stands for **Stream Editor**.

It is used to **read, modify, and transform text automatically** without opening a file manually.

👉 Think of Sed as:
"Find something → change it → print result"

🔹 How Sed Works

- Reads input line by line
- Applies commands
- Prints output (by default)

🔹 Basic Syntax

Sed 'command' filename

Example: Sed 's/apple/orange/' file.txt

👉 It Replaces first "apple" with "orange" in each line

🔹 When to Use Sed

- Replace text
- Delete lines
- Automate editing
- Quick file modifications

🔹 Common Sed Commands

1. Substitute (replace) - Sed 's/old/new/' file.txt
2. Replace in specific line - Sed '2s/apple/orange/' file.txt
3. Delete lines - Sed '3d' file.txt (Deletes line 3)
4. Print specific lines - Sed -n '2p' file.txt (Prints only line 2)
5. Insert line - Sed '2i Hello' file.txt (Inserts "Hello" before line 2)
6. Append line - Sed '2a Hello' file.txt

2) What is awk?

awk is a **powerful programming language for text processing and data extraction**.

👉 Think of awk as:
"Read structured data → process columns → print result"

🔹 How awk Works

- Reads file line by line
- Splits each line into **fields (columns)**
- Performs actions

🔹 Basic Syntax

```
awk 'pattern { action }' file.txt
```

🔹 Default Variables in awk

- $0 – Whole line
- $1 – First Column
- $2 – Second Column
- NF – Number of Fields
- NR – Line Number

🔹 Basic Examples

1. Print entire file - awk '{print}' file.txt
2. Print specific column - awk '{print $1}' file.txt
3. Print multiple columns - awk '{print $1, $3}' file.txt
4. Print line number - awk '{print NR, $0}' file.txt

🔹 When to Use awk

- Work with structured data (tables, CSV)
- Extract columns
- Perform calculations
- Generate reports

✅ Summary

- **sed** → Edit text (replace, delete) , Use sed for quick edits
- **awk** → Analyze data (columns, calculations), Use awk for logic & calculations